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Week 2026
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Session :26207

SIM-Based Secure Connectivity for Utility Operations in Non-MPLS Feasible Areas

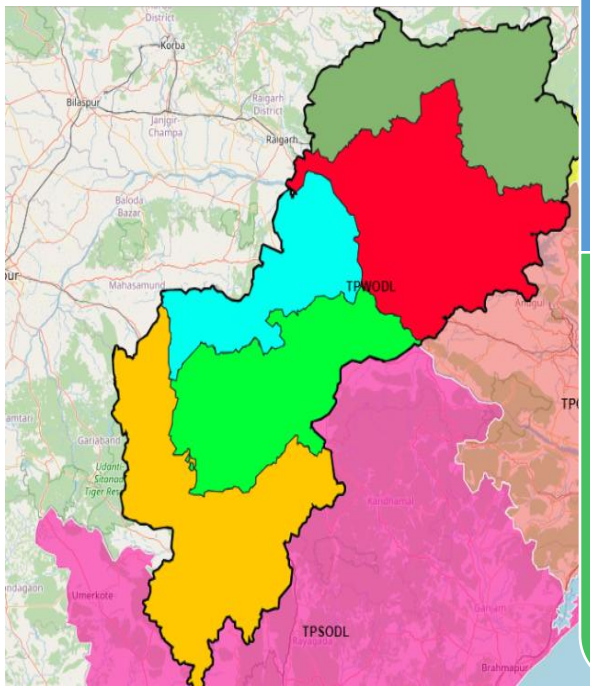
Presented By

Pritesh Kumar Srivastava, HoG Communication TP Western Odisha Distribution Limited

PRESENTATION GUIDELINES

- **Presentation Time: The total time for presentation is 12 mins. You are requested to be present on-time as per your session slot.**
- Presentation Format: All the presenting speakers are required to give a presentation as per the PPT Format. (Maximum 7 Slides). Please note that the additional presentation slides cannot be added in the shared PPT format.

INTRODUCTION



TPWODL

- 88 lakh Population Served
- 48,373 Sq. Kms of Distribution Area
- 21.42 lakh Customer Base
- Reliability improvement is one of the key KPI

Responsible for end-to-end network connectivity for both IT Offices & PSS locations through MPLS /4G/OFC

Establish reliable, secure and stable network connectivity to APSCC

Provide network connectivity to Smart Metering solution through IoT & MPLS

Provides Integrated Communication Network (ICN) connectivity with other three DISCOMs through OPGW

Communication department is backbone for the SCADA & IT Connectivity across all the offices

The entire success of remote operations of PSS depends on the QoS of communication network

CONTEXT

Overcoming Infrastructure Constraints

- MPLS not feasible everywhere due to geography, or accessibility.
- SIM-based communication fills this critical gap effectively.
- ISP Redundancy / Network Redundancy

Operational Continuity

- Ensures uninterrupted SCADA operations even in remote or infrastructure-challenged areas.
- Automatic failover from MPLS to SIM without any human intervention.

Improved Safety and Grid Stability

- Continuous SCADA communication enables faster decision-making and quicker response to network faults.
- Enhances monitoring and control of field assets, improving grid reliability.

Strengthened Cybersecurity

- Secure tunnels between field routers and head-end ensure data confidentiality and integrity.

IMPROVED NETWORK UPTIME-MORE THAN 99.5%.

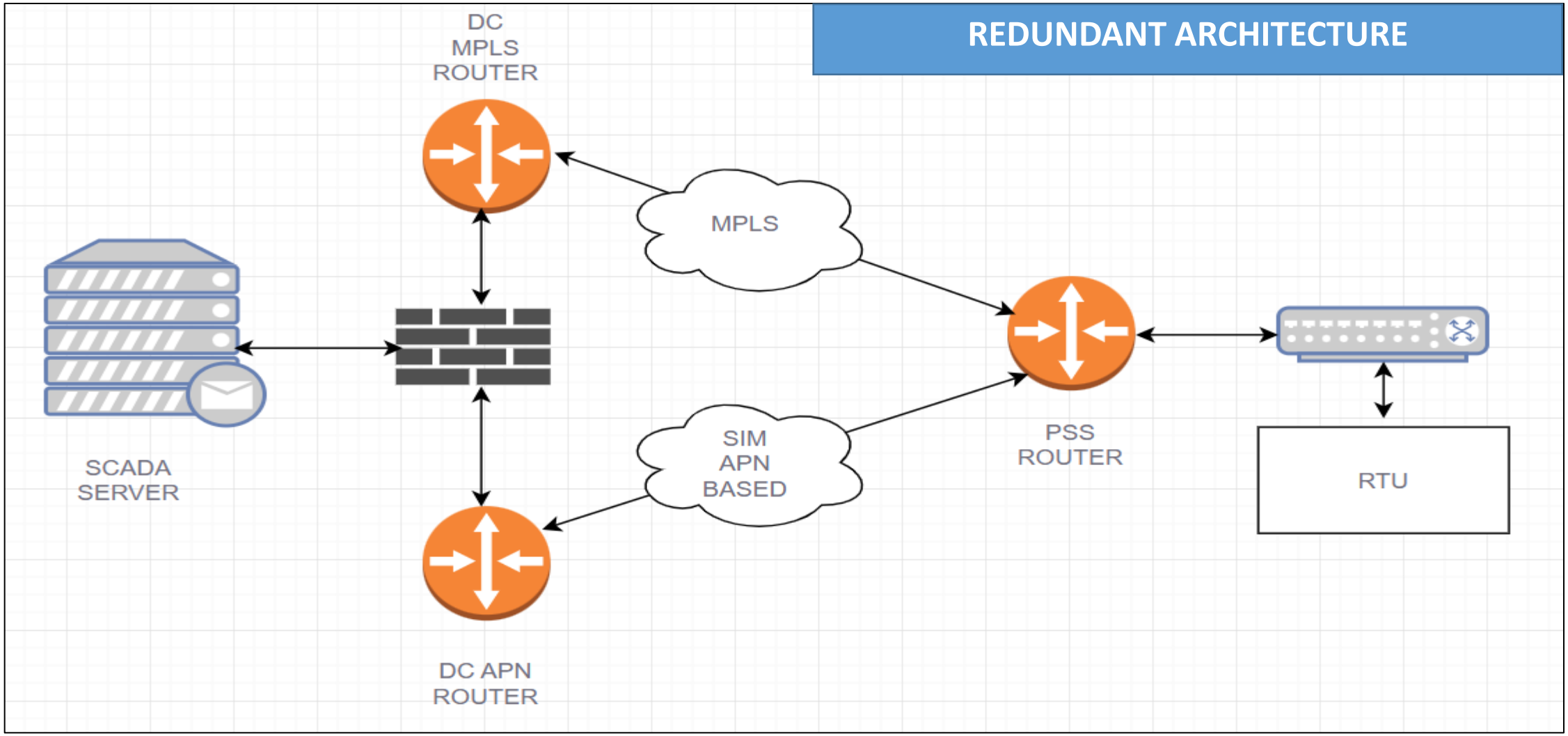
SCADA AVAILABILITY: MORE THAN 99.5%.

24*7 MONITORING OF THE GRID FOR PSCC TEAM WITHOUT INTERUPTIONS.

RELEVANCE

Geographical Scalability	Replicability Across Utilities	Cost Optimization	Technology Scalability	Future-Ready Architecture
• Easily scalable to any remote Sub station or area with SIM connectivity.	• Applicable as a model for other DISCOMs facing challenges in MPLS rollout. • Standardized design enables quick replication with minimal customization.	• The Project can be rolled out with minimal cost.	• Supports integration of additional devices such as CCTV through the router.	• Supports migration to 5G or other advanced communication technologies. • Seamless scaling for increased bandwidth and additional for SCADA/IT applications.

PRESENTATION ON THE TOPIC (1/2) (1 Min)



PRESENTATION ON THE TOPIC (2/2) (1 Min)

Optimizing Cost and Resources

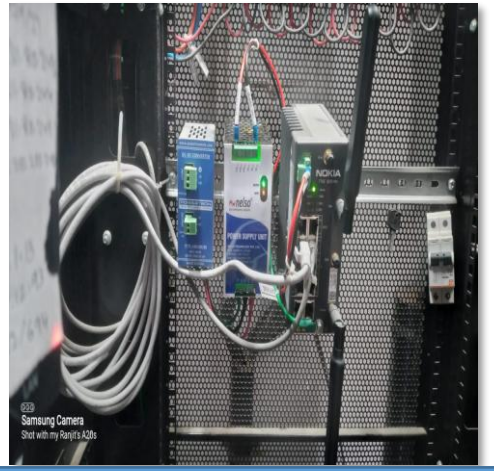
- Reduces the high capital investment associated with communication networks in non Feasible areas.
- Offers a flexible and cost-effective alternative without compromising on performance.

Enhancing Operational Resilience

- Dual communication (MPLS + SIM) strengthens the robustness of the network.
- Provides a scalable, secure, and future-oriented communication architecture.

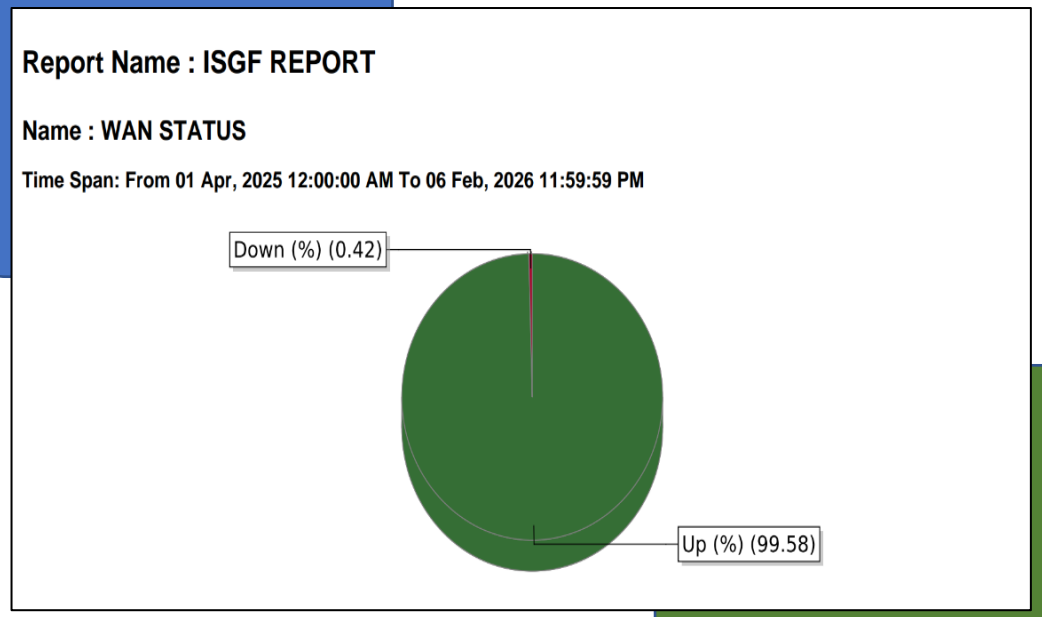
Total Cost

- Rs 108 for 1 year - per site
- Rs 37800 per year - for 350 PSS across TPWODL
- No additional investment is required for the project

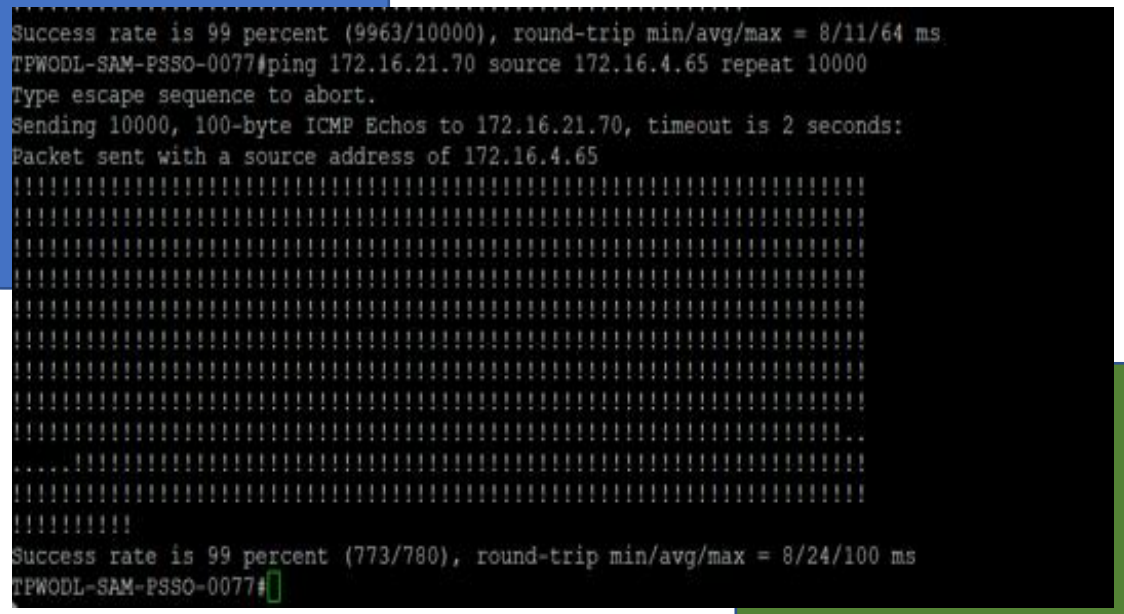


FIELD VISIT DURING INSTALLATION OF SIM

USE CASE / CASE STUDY



UPTIME REPORT- NMS



TEST RESULT

SITE HAVING REDUNDANT ARCHITECTURE HAS AN UPTIME MORE THAN 99.5 %

AUTOMATIC FAILOVER FROM MPLS TO SIM DURING MPLS OUTAGES

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THANK YOU

For discussions/suggestions/queries email: isuw@isuw.in

www.isuw.in

Links/References (If any)